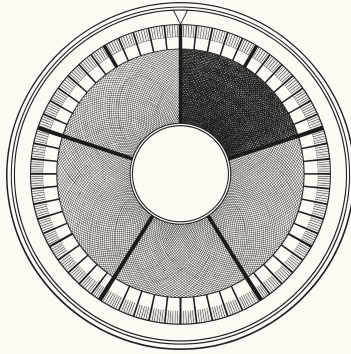




A L M A 40 : 8

Time Only Is Measured Unto Men

A calendar with no Earth in it, read from Latter-day Saint scripture

Vladimir Ulogov

No doctrine is argued true here, and none is argued false.

Why this document has that title

The title I first wrote for this was **My knowledge of God is limited, but I do have a scientific timeline that can fit him**. It is a good sentence and I could not use it, because the second half is exactly the claim this project spends two hundred pages refusing.

No timeline fits God. Not this one, not a better one, not one built with more care. The reason is not that the instrument is crude — it is that an interval-measure measures interval, and the created order is where interval is. Part IV puts the argument in the words of a fourth-century bishop who made it better than I can.

So the title is Alma's instead.

ALMA 40:8

All is as one day with God, and time only is measured unto men.

That sentence does two things at once, and both are what this project needs. It says measuring time is a **creaturely** activity — the thing given to men to do. And it says the divine mode is not that, and is not reached by doing more of it.

A calendar that counts ticks is doing the thing in the first half of that verse. It has nothing whatever to say about the second, and this document is partly an account of how the software was built so that it **cannot** say anything about the second, even by accident.

What this is and is not

A technical document with scripture in it, written for readers who know the scripture better than its author does. It reports where a piece of software and Latter-day Saint

doctrine converge and where they collide, at comparable length. The convergences are the easy half; the collisions are why it exists.

THE RULE THIS DOCUMENT WORKS UNDER

No doctrine is argued true here, and none is argued false. The software is evidence for nothing about the world.

Where the two agree, that is reported as a convergence — not as the doctrine being vindicated by a program, which would be worthless, and not as the program being vindicated by the doctrine, which would be worse.

Where they collide, the collision is stated as the **project's** problem. It is the project that has to choose, and in every case in Part V it has chosen without arguing for the choice.

PART 2

The irritation, and what was built

Read almost any account of the early universe and you find a sentence like **recombination occurred about 380,000 years after the Big Bang**.

A year is the time Earth takes to circle the Sun — definitionally, not approximately. So an event 13.8 billion years old is described in units defined by the motion of a planet that would not exist for another nine billion years.

The number is correct. It carries a passenger. `ucal` is what you get if you refuse the passenger.

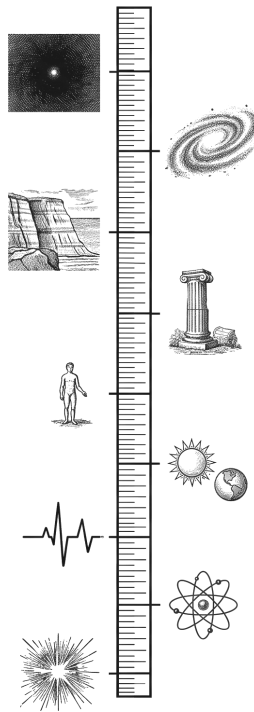
The tick and the ladder

TERM Tick

The Planck time, about 5.391×10^{-44} seconds — built from the gravitational constant, the reduced Planck constant and the speed of light. Every quantity in the system is an unsigned integer count of these.

Ticks group into **tiers**: the powers 5^{5k} , each exactly five base-5 digits. The reference rung is the **beat**, 5^{60} ticks, about 46.762 milliseconds — base five because $5^5 = 3125$ is a number of five base-5 digits, and exponent 60 because it puts the reference rung where a human can notice a duration. Those are the only two reasons, and the specification forbids finding any others.

The cost of leaving Earth units behind is that nothing on the ladder is near anything familiar. One second is 21.385061835 beats.



The domain, logarithmic: Planck tick to a single point of light in emptiness. The ceiling is about 2.29×10^{103} years; the present epoch sits at 6×10^{-94} of it.

Three commitments

Time is unsigned. The domain begins at the datum, and no earlier instant is representable — a result that would be earlier is `UCAL-E0020`, an error rather than a negative number. That is a limit on what the system can **date**, not a claim about what exists.

No floating point, anywhere. Not in a signature, a field, an intermediate, or the printed output. A lint fails the build on any float token. Cosmology is therefore done by certified interval arithmetic: it returns two numbers and a proof the answer lies between them, rather than one number and a guess.

Earth enters at one declared boundary. A single exact constant, `SECOND`, converts between ticks and seconds. Conversion in is multiplication and never rounds. The dependency is not eliminated — that is not available to anyone — it is **localised** somewhere you can point at.

PART 3

Where the doctrine got there first

Four places, and the first is not a resemblance.

Reckoning is according to the planet

The mechanism that produces calendars in this system has one rule above the others: there is **one** derivation path, every body goes through it, and Earth is an ordinary instance rather than the template. No Earth branch, no crate named after a body, and a test that builds Earth's calendar and Mars's by the identical code from data alone.

That rule exists to prevent a specific failure: Earth quietly becoming the template, one convenience at a time, until the mechanism is an Earth calendar with parameters.

DOCTRINE AND COVENANTS 130:4-5

In answer to the question — Is not the reckoning of God's time, angel's time, prophet's time, and man's time, according to the planet on which they reside? I answer, Yes.

That is the rule's premise, stated in 1843, without the software.

Not that Earth's reckoning is primary and others approximate it. Not that there is a true reckoning somewhere that local ones are shadows of. **Reckoning is indexed to a body**, and the indexing is the whole story.

WHAT THE CONVERGENCE IS WORTH

It is the most direct statement of this premise found in any tradition surveyed, and it predates the software by 183 years.

What it does not supply is the mechanism. “According to the planet on which they reside” is a proposition; expanding a continued fraction over two orbital periods is a procedure. The convergence is at the level of the claim, and the document should not inflate it past that.

A conversion with both frames named

Every foreign unit in this system crosses one declared boundary, and a bridge constant is required to say what it converts **between** — both sides, tagged.

ABRAHAM 3:4

And the Lord said unto me, by the Urim and Thummim, that Kolob was after the manner of the Lord, according to its times and seasons in the revolutions thereof; that one revolution was a day unto the Lord, after his manner of reckoning, it being one thousand years according to the time appointed unto that whereon thou standest.

Read as a conversion statement, it has the shape the specification requires:

in the verse	what it corresponds to
one revolution = one day unto the Lord	the source quantity, in source units
“after his manner of reckoning”	the source profile tag
one thousand years	the target quantity
“according to the time appointed unto that whereon thou standest”	the target profile tag

The verse does not say “a day is a thousand years” and leave you to work out whose day and whose years. It names both frames.

That is worth pausing on, because ordinary technical writing routinely fails at it. **The timeout is 30** — thirty what, measured by whom? A ratio with only one frame named is not a conversion; it is a number waiting to be misread. Whatever one thinks of the source, the **form** here is the form the specification demands.

Law with bounds and conditions

Every body parameter in the system carries an epoch, a rate of secular change, and a **window of validity**. Inside the window it computes; outside it warns rather than extrapolating confidently.

Earth's rotation is lengthening by about 1.8 milliseconds per century, and its tropical year shortening by about half a second. A parameter treated as a constant is wrong the moment you leave the epoch it was measured at — which matters most at exactly the deep-time scale this project targets.

DOCTRINE AND COVENANTS 88:38

And unto every kingdom is given a law; and unto every law there are certain bounds also and conditions.

A regularity holds **within bounds and conditions**, and the bounds are part of the law rather than a caveat attached to it. That is the design decision, stated as a general proposition about law.

What can be numbered, and by whom

This is the one that gave the document its title, and it does the most work.

MOSES 1:37

And the Lord God spake unto Moses, saying: The heavens, they are many, and they cannot be numbered unto man; but they are numbered unto me, for they are mine.

Notice precisely what is said to be beyond numbering: the **heavens**. Not durations. Not intervals. Not the count of ticks between two moments in the created order.

And beside it, Alma:

ALMA 40:8

Now whether there is more than one time appointed for men to rise it mattereth not; for all do not die at once, and this mattereth not; all is as one day with God, and time only is measured unto men.

Measuring time is presented as what creatures do, in contrast to a divine mode where all is as one day. So this tradition, like several others surveyed, locates time-measurement firmly on the creaturely side — and treats that as its proper place rather than as a limitation to be transcended.

A calendar that counts ticks is doing the thing that is given to men to do. That is a modest claim, and it is the one this project has been circling since its first page.

PART 4

The instrument's own limit

Before the collisions, the thing the two agree about most deeply — and the reason the title had to change.

What the instrument measures

Gregory of Nyssa, in the fourth century, argued against Eunomius that **διάστημα** — interval, extension, the spread between before and after — is the mark of createdness. Everything created is διαστηματικός; what is uncreated is ἀδιάστατος, without interval. The gap is not one of degree along a scale. It is a difference of **kind**: there is no interval there for a measure to be a measure of.

Every quantity in this system is an interval. The tick count is the interval from the datum. A duration is an interval. An uncertainty window is an interval with its bounds carried. The whole apparatus, from the 512-bit integer to the cosmological enclosures, measures διάστημα and nothing else.

WHICH IS WHY NO TIMELINE CAN FIT GOD

Not because the timeline is imprecise, and not because a better one might manage it. Adding digits does not approach what has no interval, in the same way that no finite magnitude approaches the infinite by growing.

The instrument reaches exactly as far as interval reaches. That is not a confession of failure. It is a statement about what an interval-measure **is** — and it is the same point Alma makes in five words.

And what it refuses to compute with

The design's central move follows from that. The datum — tick 0 — is **stipulated**: declared, not measured. The published age of the universe is 13.787 ± 0.020 Gyr, and that uncertainty is a fifty-eight-digit number of ticks, about 0.145% of the whole span. An exact count cannot inherit it.

So the physical claim about where the origin falls is kept **separately**: cited, with its exact magnitude, in a type that has **no arithmetic operations at all**. You can read it, print it, cite it. You cannot compute with it — and three tests exist whose job is to **fail to build** if anyone ever does.

● ● ● *tests/compile_fail/signed_window_as_operand.rs*

```
use ucal_core::{Instant, Profile, UC1};

fn main() {
    let t: Instant<UC1> = Instant::zero();
    let claim = UC1::big_bang_claim();
    // A SignedWindow is not a Delta and must never become one.
    let _ = t.checked_add(&claim);
}
```

That is the whole of the project's contribution: not the observation that an instrument should declare what it cannot reach — which is old, and better said by Cusanus in 1440 — but that the declaration can be **enforced by a machine** rather than left to the author remembering.

PART 5

Where they collide

Four, and one that looks like a collision and is not. In every case the problem is the project's rather than the doctrine's, and in every case the project has chosen without arguing for the choice.

The one that is not a collision: the datum

This is where a reader would most expect the two to come apart, and the title of this document is why they do not.

`ucal`'s profile UC-1 stipulates tick 0 and conventionally identifies it with the FLRW $t \rightarrow 0$ limit. The domain is unsigned: no earlier instant is representable, and asking for one is an error rather than a negative number.

It is easy to read that as **nothing existed before** — and I wrote an earlier draft of this document that read it exactly so, and built a collision on top of it. That was wrong, and the specification says so plainly. `BIG_BANG_CLAIM` is a **signed** window precisely because the limit “may lie before the datum, which is not representable as a tick.” A system asserting that nothing precedes the datum would have no use for a type that can express something preceding it.

WHAT THE UNSIGNED DOMAIN ACTUALLY SAYS

The datum is the best available starting point for time **as this system can date it**, and everything it dates is dated from there. What may lie earlier is a question the instrument declines rather than answers.

It is a limit on **range**, not a claim about **existence**. Alma's verse is the whole of it: time is measured unto men, and a measuring instrument reaches as far as measuring reaches.

DOCTRINE AND COVENANTS 93:29

Man was also in the beginning with God. Intelligence, or the light of truth, was not created or made, neither indeed can be.

Taken with the King Follett discourse of 1844 — in which the word rendered **created** is argued to mean **organise**, and matter is held co-eternal with God — the picture is not one in which time began, but one in which a particular organisation began, out of materials that did not.

The instrument does not contradict that. It is **silent** about it, and silence is not contradiction. Where the doctrine speaks of what precedes organisation, the instrument has nothing to say, because there is nothing there it can date.

What remains, which is a limitation and not a disagreement

If the doctrine holds there is a dateable “before” — an eternity of organised matter preceding this arrangement — then the instrument’s **reach** falls short of the doctrine’s **scope**. That is a real limitation and worth naming as one.

The specification anticipates it. A second profile, UC-Θ, would place the datum at organisation rather than at a physical origin and make the interval before it representable. Under UC-Θ the unsigned domain becomes **room** rather than an edge. It has never been built.

So the honest statement is narrow: UC-1 cannot date what precedes its datum, and UC-Θ would extend the range rather than correct a false claim. Whether extending it is desirable is a separate question, and one this document does not settle.

AN INVERSION WORTH STATING ANYWAY

The book this document summarises reads nine traditions. Judged by the Patristic and Latin material — Augustine, Basil, Aquinas, all holding creation **ex nihilo** including time — UC-Θ is the **heterodox** profile: it posits pre-existing material shaped rather than made.

Read from Latter-day Saint scripture, that is reversed. UC-Θ is the natural profile and UC-1 the foreign one.

Two profiles here are not two configurations of one system. They are two cosmologies — and the project ships the one that reaches less far for this reader, without having argued that it should.

1. Kolob is a body; the ladder is a grid

The mechanism permits no privileged body. The universal ladder is the powers 5^{5k} — an abstract grid — and the specification is explicit that no body's period may occupy a special place in it.

ABRAHAM 3:9

And thus there shall be the reckoning of the time of one planet above another, until thou come nigh unto Kolob ... which Kolob is set nigh unto the throne of God, to govern all those planets which belong to the same order as that upon which thou standest.

Abraham 3 describes a hierarchy of reckonings, and the hierarchy has a **place** at its centre. The governing referent is a body.

WHERE THEY COLLIDE

Both arrangements are hierarchies with a privileged referent. They differ in what sort of thing sits at the privileged point: a named body, or a power-of-5 grid.

So the honest description of what this project did is not that it abolished the privilege. It **relocated** it — from a body to a grid.

And a grid is not neutral merely by being abstract. It was chosen: base five, anchored at 5^{60} , with the reference rung placed where a human can notice a duration. A reader of Abraham 3 could fairly say that a ladder anchored to human perception is parochial about something too — just about a different thing.

I think the relocation is defensible, and the reason is that a power ladder can be re-anchored without changing any arithmetic, while a governing body cannot be replaced without changing the cosmology. But that is an argument the project should make rather than assume, and until it was written down here it had assumed it.

2. Revelation has no place in the schema

Every body parameter in the system records **how it was determined**: measured to a stated precision, derived from other parameters, or cited to a published source with an uncertainty window. The field is closed; those are the values it admits.

A revealed ratio fits none of them.

The consequence is precise and worth stating exactly, because it cuts one way and not the other. Abraham 3:4 **can** enter the system as a declared bridge constant, because a bridge is required to be declared and tagged on both sides and the verse is. It **cannot** enter as an anchor — the constant that fixes where a calendar's count begins in phase — because an anchor must carry a determination method and a stated uncertainty, and no reading of the text supplies either without inventing them.

WHERE THEY COLLIDE

The system is hospitable to this tradition's **ratios** and closed to its **epistemology**.

That is a real exclusion and it is not neutral. It is not that revelation is judged unreliable; the schema has no way to represent the category at all, and a schema that cannot represent a category has taken a position on it by omission.

3. A revealed ratio does not land on the grid

A smaller finding, reported because an attentive reader would otherwise wonder, and because it says nothing at all about the text.

Take Abraham 3:4 as a bridge constant and convert it. One Kolob revolution is 1,000 Julian years, which is 585,348,807,057,053,493,600 followed by thirty-three zeros of ticks — and that is **not** a whole number of beats. The remainder is 35,632,189,513,851,911,760,866,641,998,291,015,625,000 ticks.

The reason is arithmetic and it has nothing to do with Kolob. A thousand Julian years is a whole number of **seconds**; the second carries thirty factors of five, while the beat carries sixty. Any quantity stated in years misses the tier grid by construction, from any source whatever.

WHY THIS IS IN THE DOCUMENT AT ALL

Because it **looks** like a finding about scripture and is not, and a reader skimming could take it for one.

It is the same incommensurability that makes one second 21.385 beats rather than a round number, showing up somewhere nobody was looking for it.

4. Scripture chronology is classified as declared, not derived

The system distinguishes calendars whose rules it **derives** from a body's motion from calendars whose rules are **declared tables**. That distinction is factual rather than evaluative — it records where a rule's authority comes from — but it does not flatter, and it is applied without exception.

Applied to the Gregorian calendar, it produces an uncomfortable result: the leap rule of 1582, $97/400$, is not derivable from Earth's motion at any depth, and a rule twelve times simpler is more accurate. The Julian rule **is** derivable — it is the first thing the mechanism produces.

DOCTRINE AND COVENANTS 77:12

... as God made the world in six days, and on the seventh day he finished his work, and sanctified it ... even so, in the beginning of the seventh thousand years will the Lord God sanctify the earth ...

WHERE THEY COLLIDE

A seven-thousand-year structure is a declared period. The mechanism does not derive it from any body's motion, because no body's motion yields it.

So the instrument would classify it exactly as it classifies the Gregorian leap rule: **declared, not derived**.

That has to be said, because a system that called the Gregorian declared and a scriptural period something else would not be classifying. It would be flattering, and a classification that bends for the author's sympathies is not a classification.

The classification is not a verdict. Chapter after chapter of the book makes the same point about the Gregorian, and the strongest defence of **declared** calendars in the whole survey comes from a different tradition entirely — the Orthodox churches that keep the Julian calendar knowingly, on the crudest rule available, for reasons of communion that have nothing to do with accuracy.

Derivation is not a calendar's only criterion. It is the only one this instrument can measure, which is a fact about the instrument.

PART 6

What is not claimed

A negative inventory, because a document with scripture in it will be quoted and the quoting will not always be careful.

Not that any doctrine is true — or false. The software is evidence for nothing about the world, and its agreements are not endorsements.

Not that the scripture anticipated this work — D&C 130:4–5 states a proposition that this project also holds. Body-relative reckoning is a thought available to anyone who considers another planet, and there is one good algorithm for continued fractions.

Not that the instrument reaches God — it measures interval, and interval is the mark of the created order. That is Part IV, and it is why the title changed.

Not that time began — the system stipulates a datum and counts from it. Whether anything began is a question it declines, and Part V is honest that its unsigned domain nevertheless leans one way.

Not that base 5 is meaningful — $5^5 = 3125$ is five base-5 digits. The specification explicitly forbids any constant acquiring significance from resembling a number in a tradition, and that rule was written before the survey began.

Not that the seven thousand years are wrong — the instrument classifies the period as declared. It has no opinion about whether a declared period is correct, and neither does its author in this document.

Not that the system is useful — no task you have needs a Planck-tick count. That is stated in the project's own preface as a fact rather than an apology.

What is claimed

A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.

The first clause is old. Cusanus has it in 1440; Gregory of Nyssa supplies the reason a millennium before that; Alma says it in eleven words.

The second is the contribution, and it is small and specific. A comment addresses a reader who is paying attention. A runtime check addresses a program that reaches a particular line. A **type** addresses everyone who ever writes code against the library, on every path, including paths nobody thought about — and it requires nothing of them at all.

Someone who thinks the distinction between a stipulated datum and a measured origin is pedantic sits down, writes the line that ignores it, and is stopped in under a second by something with no interest in whether they agree.

PART 7

Where to go from here

if you want	read
the software	<code>cargo install ucal</code> , or <code>cargo add ucal-core</code>
the chapter this document draws on	Life, the Universe, and God , chapter 23 — the same material at greater length, alongside eight other traditions
the datum argument in full	the same book, Part IV — four chapters
what the rules mean	<code>spec/RULES.md</code> — all twenty-four, and what enforces each
a shorter general account	<code>UCAL_INTRO</code> — 29 pages, no scripture

One last note, and it is the note the book itself ends on.

The type system stops the claim about the origin from entering the arithmetic. It has no reach whatever over what a person **reads**. A reader who looks at a sixty-one-digit integer will take it for a fact about being, because that is what a very precise number looks like — and no type signature stands between a number and its reader.

The instrument contains that; it does not cure it. Which is, in the end, a narrower version of the thing Alma said: the measuring is ours to do, and what it does not reach is not reached by measuring more carefully.

Tick zero is a stipulated reference point, conventionally identified with the $FLRW\ t \rightarrow 0$ limit. It is not a measurement and not an observed event.

TIME ONLY IS MEASURED UNTO MEN

– printed by `ucal datum` on every run